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Skew-Symmetric Error Distribution Analysis of Elasticity Dynamics of Firm Financial Statement Components: A Case Study in Indonesian Stock Markets using The Cobb-Douglas Linearized Log Form

Abstract

This study investigates the elasticity of total assets and number of employees on total revenue using a Cobb-Douglas production framework on Indonesian public firms. It compares classical OLS regression with a skew-symmetric regression model using the BFGS optimization algorithm. Findings show the skew-symmetric model outperforms classical models in terms of predictive power, residual normality, and information criteria (AIC/BIC), providing more accurate estimations in contexts with asymmetric data.

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1.Introduction

Economics and Statistical Modelling

- Statistical modeling has become central to financial statement analysis, with a **shift toward advanced econometrics** (e.g., time series, regression) for understanding firm behavior and valuation (Brooks, Damodaran) [1] [2].

Modelling Economic Phenomenon

- Techniques like the **Cobb-Douglas Production Function** are used to model firm **output** (Total Revenue) via **capital** (Total Assets) and **labor** (Total Employees), typically log-transformed to estimate elasticities in a regression framework, with residuals following a **normal distribution process**.

The linear regression's normality assumption

- Traditional regression models assume **normality with logarithmic quantity emphasis**, but real-world financial data often exhibit skewness, especially in volatile emerging markets like Indonesia. [3]

The skew symmetric distribution's role

- **Skew-Symmetric Distributions** (Azzalini; Arellano & Bover) allow for better modeling of asymmetric residuals and improve estimation robustness under non-normal data conditions. [4]. Prior studies like Jiminchi et.al also confirms this [5]

- This study **applies both classical and skew-symmetric Cobb-Douglas models** to Indonesian firms, testing which better captures productivity through statistical diagnostics and normality/fit tests.

References:

1. Brooks, C. (2014). Introductory econometrics for finance (4th ed.). Cambridge University Press. (p. xxix).
2. Damodaran, A. (2012). Investment valuation: Tools and techniques for determining the value of any asset. John Wiley & Sons. (p. 36).
3. Bank Indonesia. (2021). Financial stability review: Safeguarding economic recovery amid financial sector resilience. Bank Indonesia.
4. Azzalini, A. (1985). A class of distributions which includes the normal ones. Scandinavian Journal of Statistics.
5. Jiminchi, M., Kawasaki, Y., Miyamoto, D., Saka, C., & Nagata, S. (2023). Statistical Modelling of Financial Data with Skew-Symmetric Error Distributions. Symmetry, 15(9), 1772.

2. Literature Review

- **Theories Used:**
 - Theory of Production (Cobb & Douglas);
 - Theory of the Firm (Coase); and
 - Classical regression limitations addressed using Azzalini’s skew-normal distribution.
- **Framework and Hypothesis:**
 - **H1** : Variables of Total Assets and Total Employees correlate significantly to a firm’s Total Revenue.
 - **H2**: A regression model incorporating skew-symmetric distributions provides a better estimation of firm performance compared to traditional symmetric regression models.

Theory of Production

- The firm combines and reconciliates the inputs of labor and capital into production.
- Labor and capital have their tradeoffs between one and another in respective elasticities, by which one can utilize or prefer one or the other [1].
- In financial statement analysis, the Cobb-Douglas function serves as a framework to assess how efficiently a firm allocates resources.
- Empirical studies have validated this function in corporate finance, showing that asset utilization and workforce management significantly impact firm revenue such as the study of Hall and Jones [2]

Theory of The Firm

- According to Coase, firms exist to minimize transactions costs and operate efficiently in response to market forces and aim to maximize profits subject to competitive pressures [3].
- In financial statement analysis, this theory underpins key financial ratios, such as return on assets (ROA), return on equity (ROE), and revenue efficiency metrics, to evaluate firm performance [4].

References:

- 1.Cobb, C. W., & Douglas, P. H. (1928). A theory of production. American Economic Review, 18(1), 139-165.
- 2.Hall, R. E., & Jones, C. I. (1999). Why do some countries produce so much more output per worker than others? Quarterly Journal of Economics, 114(1), 83-116.
- 3.Coase, R. H. (1937). The nature of the firm. Economica, 4(16), 386-405.
- 4.The nature of the firm. Economica, 4(16), 386-405.

3. Methodology and Statistical formulation

The Outline of Equations

The Fundamental Formula

1.

$$R = \kappa A^{\beta_1} N^{\beta_2} \varepsilon$$

Fundamental formula is built upon the model proposed by Charles Cobb and Paul Douglas, in which Output is the result.

- R : Revenue
- κ : Parameter
- A : Assets
- N : Number of employees
- ε : Error term
- β_i : Elasticities

The Classical Regression

2.

$$\log R = \beta_0 + \beta_1 \log A + \beta_2 \log N + \epsilon$$
$$\epsilon \sim \mathcal{LN}(0,1)$$

The formula regressed is a log form Cobb - Douglas, by which it is the base model taken its logs. The residuals follow a Classical Regression process of Log-Normal Distributions

- $\beta_0 = \log \kappa$
- $\epsilon = \log \varepsilon$
- $\mathcal{LN}$: Log-Normal distribution

The Skew-Symmetric model

3.

$$f(\epsilon) = \frac{2}{\omega} \phi\left(\frac{\epsilon - \zeta}{\omega}\right) \Phi\left(\alpha \frac{\epsilon - \zeta}{\omega}\right)$$

We optimize the parameters of the Classical Regression so that the residual instead of following a Log-Normal Distribution, follows a Skew-Symmetric Distribution. Optimization is via BFGS Algorithm

- ω : Scaling parameter ($\omega > 0$)
- ζ : Location parameter
- α : Skewness parameter
- ϕ, Φ : PDF and CDF resp.

The Skew-Symmetric regression

4.

$$\log R = \alpha_0 + \alpha_1 \log A + \alpha_2 \log N + \epsilon$$
$$\epsilon \sim f(\epsilon|\zeta, \omega, \alpha)$$

We regressed a new model with new parameters α (alpha) where the coefficients are adjusted to the residuals of the Skew-Symmetric distribution

Coef: $\beta_i \rightarrow \alpha_i$; not to be confused with α as a skewness parameter

Properties of Cobb-Douglas

Diminishing Marginal Property

1.

Marginal Product of Capital : $R_A = \beta_1 \frac{A}{R} > 0$; $R_{AA} = -\beta_1 \frac{A}{R^2}$

Marginal Product of Labor : $R_N = \beta_2 \frac{N}{R} > 0$; $R_{NN} = -\beta_2 \frac{N}{R^2}$

2.

Returns to scale

Constant : $\beta_1 + \beta_2 = 1 \Rightarrow R(kx) = kR(x)$

Increasing : $\beta_1 + \beta_2 > 1 \Rightarrow R(kx) > kR(x)$

Decreasing : $\beta_1 + \beta_2 < 1 \Rightarrow R(kx) < kR(x)$

* $\forall x = (A, N)$

Marginal Rate of Technical Substitution (MRTS)

3.

$$MRTS = \frac{R_A}{R_N} = \frac{\beta_1 A}{\beta_2 N}$$

Methods

- **Type:** Quantitative secondary data using yfinance (YahooFinance)
- **Sample:** 90 sample of IDX-listed firms (Q3 2023–Q3 2025)
- **Regression Models:**
 - Classical OLS (log-linear)
 - Skew-symmetric with BFGS optimization
- **Tests:**
 - Shapiro-Wilk (normality)
 - Breusch-Pagan (heteroscedasticity)
 - Durbin-Watson (autocorrelation)
 - AIC/BIC (model quality)

3. Methodology and Statistical formulation

The Skew-Symmetric Distribution

1.

$$L(\theta) = \prod_{i=0}^n \left(\frac{2}{\omega}\right)^n \phi(z_i) \Phi(\alpha z_i)$$

- The regression of the log-linear function is analogous as standard linear regression

$$y_i = a + bx_i + \epsilon_i \quad (1)$$

- The log-likelihood of the function is formulated in the form :

$$l(\epsilon|\theta) = \sum_i \log f(\epsilon|\theta) \quad (2)$$

- Now we insert the x_i as ϵ and $\theta = (\zeta, \omega, \alpha)$ and also Φ, ϕ as the CDF and PDF respectively, thus the skew symmetric likelihood is:

$$l(\epsilon|\theta) = \sum_i \log \left[\phi\left(\frac{\epsilon - \zeta}{\omega}\right) \Phi\left(\alpha \frac{\epsilon - \zeta}{\omega}\right) \frac{2}{\omega} \right] \quad (3)$$

- Or:

$$l(\epsilon|\theta) = \sum_i \log(2) - \log(\omega) + \log \left[\phi\left(\frac{\epsilon - \zeta}{\omega}\right) \right] + \log \left[\Phi\left(\alpha \frac{\epsilon - \zeta}{\omega}\right) \right]$$

- The normalization factor of $z = (\epsilon - \zeta)/\omega$:

$$\int_{-\infty}^{\infty} f(\epsilon; \zeta, \omega, \alpha) d\epsilon = \frac{2}{\omega} \int_{-\infty}^{\infty} \phi(z) \Phi(\alpha z) \omega dz = 2 \int_{-\infty}^{\infty} \phi(z) \Phi(\alpha z) dz = 1 \quad (4)$$

$$\therefore \int_{-\infty}^{\infty} \phi(z) \Phi(\alpha z) dz = \frac{1}{2}$$

- Since $\phi(z)$ follows a Gaussian distribution of $N(0,1) \rightarrow \phi(z) = \int e^{-\frac{z^2}{2}} dz$, for z_i :

$$z_i = \frac{\epsilon_i - \zeta}{\omega} = \frac{y_i - a - bx_i - \zeta}{\omega} \quad (5)$$

- The final log-likelihood becomes :

$$\mathcal{L}(\theta) = -n \log(\omega) - \frac{n}{2} \log(2\pi) - \frac{1}{2} \sum_{i=1}^n z_i^2 + \sum_{i=1}^n \log \Phi(\alpha z_i) + n \log(2) \quad (6a)$$

- Or in exponential form

$$L(\theta) = \prod_{i=0}^n \left(\frac{2}{\omega}\right)^n \phi(z_i) \Phi(\alpha z_i) \quad (6b)$$

The Hessian

2.

$$H_{k+1} = (I - \rho_k s_k y_k^T) H_k (I - \rho_k y_k s_k^T) + \rho_k s_k s_k^T$$

- The Hessian matrix is:

$$H = \nabla^2 \mathcal{L} = \begin{bmatrix} \partial_{\theta_1 \theta_1} \mathcal{L} & \cdots & \partial_{\theta_1 \theta_k} \mathcal{L} \\ \vdots & \ddots & \vdots \\ \partial_{\theta_k \theta_1} \mathcal{L} & \cdots & \partial_{\theta_k \theta_k} \mathcal{L} \end{bmatrix} \quad (7)$$

- An example can be demonstrated as:

$$\frac{\partial^2 \mathcal{L}}{\partial \alpha^2} = - \sum_i \frac{1}{\Phi(\alpha z_i)^2} [\phi(\alpha z_i)^2 - \alpha z_i \phi(\alpha z_i) \Phi(\alpha z_i)]$$

- Since this is too tedious to compute by hand. The numerical algorithm of the BFGS comes to the rescue. The BFGS is formulated commonly by the following iteration:

- Start with initial guess and Hessian estimate θ_0 and $H_0 = I$ respectively where I is the identity matrix. At the k^{th} iteration:

$$\theta_{k+1} = \theta_k - H^{-1} \nabla \mathcal{L}(\theta_k) \quad (8)$$

- Update via:

$$s_k = \theta_{k+1} - \theta_k ; y_k = \nabla \mathcal{L}(\theta_{k+1}) - \nabla \mathcal{L}(\theta_k); \rho_k = \frac{1}{y_k^T s_k} \quad (9a)$$

$$H_{k+1} = (I - \rho_k s_k y_k^T) H_k (I - \rho_k y_k s_k^T) + \rho_k s_k s_k^T \quad (9b)$$

- The matrix is positive definite. To show this:

- We let : $V = I - \rho_k s_k y_k^T$
- So, equation (9b) is : $H_{k+1} = V H_k V^T + \rho_k s_k s_k^T$
- Basically, $V H_k V^T$ is symmetric positive semi-definite because $H_k > 0$
- Also, $\rho_k s_k s_k^T$ is positive semidefinite and adds positive curvature along s_k and for $\forall z \neq 0, z^T H_{k+1} z = (Vz)^T H_k (Vz) + \rho_k (s_k^T z)^2 \geq 0$
- So, if $z \neq 0$, we can show this is strictly positive since either $Vz \neq 0$ (the 1st term > 0) or $s_k^T z \neq 0$ (the 2nd term > 0). With $y_k^T s_k > 0$ this yields $H_{k+1} > 0$. Thus, positive definiteness is preserved

The BFGS Algorithm

3.

$$\|\nabla f_k\| > \epsilon \Rightarrow p_k = -H_k \nabla f_k$$

- The implementations used is:

$$H_{k+1} = H_k - \frac{H_k s_k s_k^T H_k}{s_k^T H_k s_k} + \frac{y_k y_k^T}{y_k^T s_k} \quad (10a)$$

- The implementations used is the inverse Hessian:

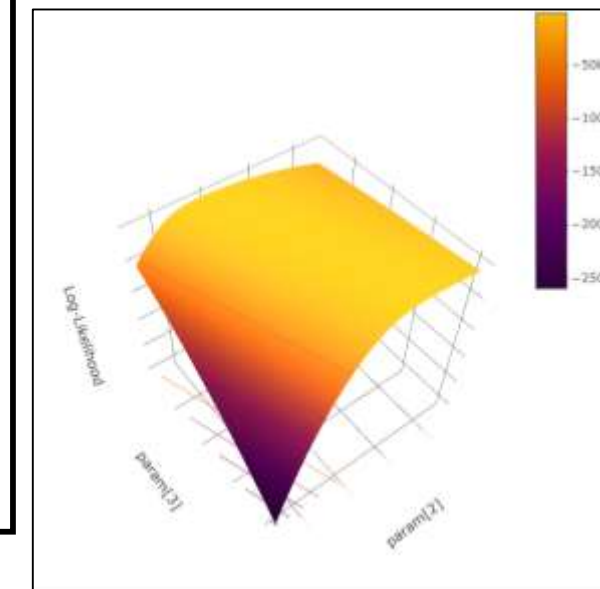
$$H_{k+1} = H_k - \left(I - \frac{s_k y_k^T}{y_k^T s_k} \right) H_k \left(I - \frac{y_k s_k^T}{y_k^T s_k} \right) + \frac{s_k s_k^T}{y_k^T s_k} \quad (10b)$$

- The algorithm typically run by R given $x_0, H_0 = I$ and repeating for $k = 0, 1, 2, \dots$:

- Given starting point x_0 , convergence tolerance $\epsilon > 0$, inverse Hessian approximation H_0 ; $k \leftarrow 0$
- While $\|\nabla f_k\| > \epsilon$, Compute the search direction $p_k = -H_k \nabla f_k$;
- Set $x_{k+1} = x_k + \alpha_k p_k$ where α_k is computed from a line search procedure to satisfy the Wolfe conditions.
- Define $s_k = x_{k+1} - x_k$ and $y_k = \nabla f_{k+1} - \nabla f_k$;
- Compute H_{k+1} via equation (10b)
- $k \leftarrow k + 1$
- end(while)

*see: Nocedal & Wright(2000), *Numerical Optimization*, Ch. 6; p.139-141

Log-Likelihood Plot



- The plot is plotting:

$$\mathcal{L} = \sum_{i=1}^n \log f_i(\theta | \theta = \alpha_1, \alpha_2)$$

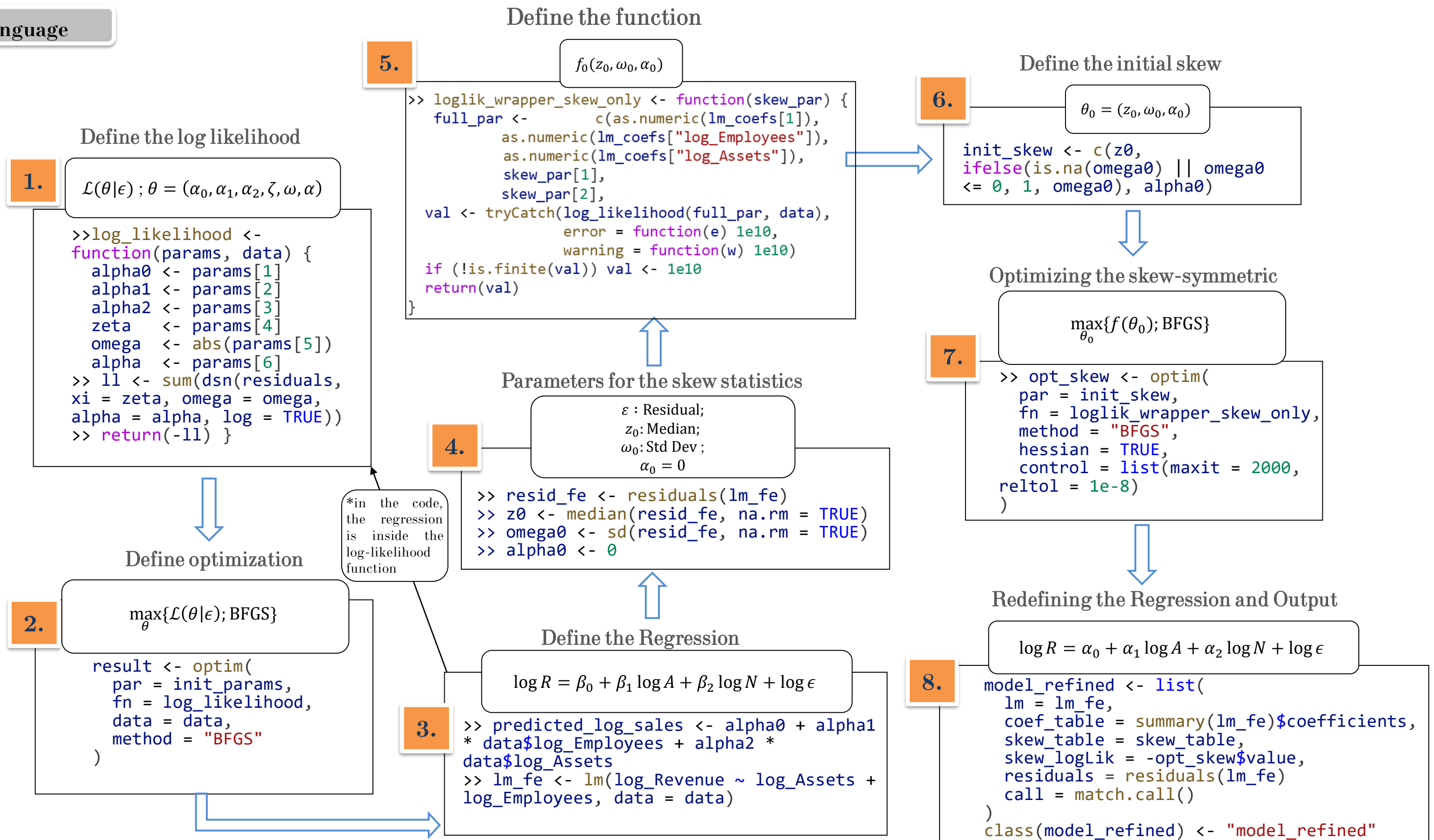
- The plotting of the likelihood function reflects decreasing probability mass as the parameters move away from the MLE
- So as ϵ_i gets bigger, $f(\epsilon_i)$ gets smaller, and the log becomes more negative

3. Methodology and Statistical formulation

Simplified Outline of Code in R-Language

*Libraries used:

- readxl – Importing Excel-based financial data
- tidyverse – Data manipulation and transformation
- lmtest – Regression diagnostics (e.g. BP,DW tests)
- sandwich – Robust variance–covariance estimation
- sn – The Skew-Normal Distribution function
- optimx – The optimization for BFGS
- car – Regression utilities and diagnostics
- ggplot2 – Data visualization (later for diagnostics plot)
- plotly – The 3D plot of Cobb-Douglas
- psych – Descriptive stats
- knitr – For readable output
- MASS – The pseudo-inverse for Heissan matrix
- gridExtra – Multipanel figure



4. Results - Stats

Descriptive Statistics

Variable	Obs	Mean	SD	Median	Trimmed	MAD	Min	Max	Range	Skew	Kurtosis	Standard Error
Log Revenue	575	29	4.38	30.74	29.54	2.21	18.2	34.4	16.24	-1.06	-0.39	0.2
Log Assets	575	31.26	4.46	32.92	31.7	2.15	21.3	37.7	16.41	-0.98	-0.38	0.2
Log Number of Employees	575	10.7	1.43	10.61	10.71	1.44	6.17	14.1	7.95	-0.11	0.09	0.1

Descriptive Statistics shows Skewness:

Based on the summary of descriptive statistics, the data shows a **heavily skewed data for the Revenue and Total Assets** of each firm

Regression

• Classical Regression Results

Variable	Estimate	Std.error	t-value	Pr(> t)	R-square	p-val
Constant	-2.05587	0.39748	-5.172	3.20e-07 ***	0.9343	< 2.2e^-16
Log Total Assets	0.90882	0.01213	74.899	< 2e-16 ***	F-stat	4064 on 2 and 572 df
Log Number of Employees	0.24789	0.03783	6.552	1.27e-10 ***		
p-val: < *0.05, < **0.01, < ***0.001						

Classical Regression predicts with good accuracy:

Shows **strong significance** in both log-assets and log-employees. R-squared = 0.9343 means 93.43% of revenue variation is explained.
Caution: Although fit is high, this model ignores skewed residuals, risking bias.

Optimized via BFGS Algorithm:

- Parameters $\alpha_0, \alpha_1, \alpha_2$, represent the regression coefficients.
- ζ, ω, α are the skew-normal distribution parameters (location, scale, skewness).

• Optimization results for the Skew-Symmetric

Parameter	α_0	α_1	α_2	ζ	ω	α
Estimate	-0.5389	0.10564	0.9699	-0.5386	1.73185	-3.1546

• The Regression results for the Skew-Symmetric model

Variable	Estimate	Std.error	t-value	Pr(> t)	R-square	p-val
Constant	-11.3699	3.467209	-3.279	0.001116 **	0.9992	< 2.2e^-16
Log Total Assets	1.280150	0.09737	0.088142	< 2e-16 ***	F-stat	5341 on 87 and 319 DF
Log Number of Employees	0.033857	0.10320	5.494	0.002572 **		
p-val: < *0.05, < **0.01, < ***0.001						

Skew-Symmetric Regression also predicts with good accuracy:

The Skew-Symmetric Regression results also show **all variables are statistically significant**. R-squared = 0.9993 indicates near-perfect model fit

Model Testing and Comparison

• Autocorrelation and Heteroskedasticity Test

Model	Breusch-Pagan Test			Durbin-Watson test
	BP Statistic	Degrees of Freedom	p-value	d-Statistic
Classical Model	306.31	88	< 2.2e-16	2.3706
Skew-Symmetric Model	186.75	87	2.99e-09	2.0871

Skew-Symmetric has low BP Test and smaller DW Test

The Breusch-Pagan test indicates heteroskedasticity in both models, **but the Skew-Symmetric model has a much lower BP statistic** (186.75 vs. 306.31). The Durbin-Watson statistic for **both models is close to 2**, indicating no autocorrelation.

• Information Criterion Tests

Metric	Classical	Skew-Symmetric Model
AIC	109.30	-428.80
BIC	475.04	-72.01

Information Criterion shows far small variabilities of deviation in the Skew-Symmetric models:

AIC and BIC are used to compare model quality. **Lower values mean better performance**. Skew-Symmetric model has much lower values (AIC = -428.80, BIC = -72.01), clearly outperforming the Classical model.

• Shapiro-Wilk Test for Normal Data

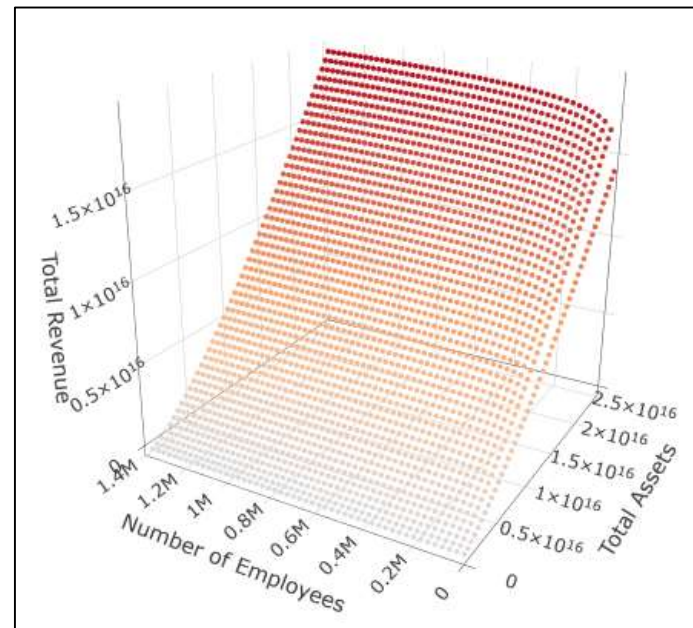
Model	W Statistic	p-value
Classical	0.77591	< 2.2e-16
Skew Symmetric	0.98251	7.85E-05

The Normality Tests shows the Skew Regression to be more Normal:

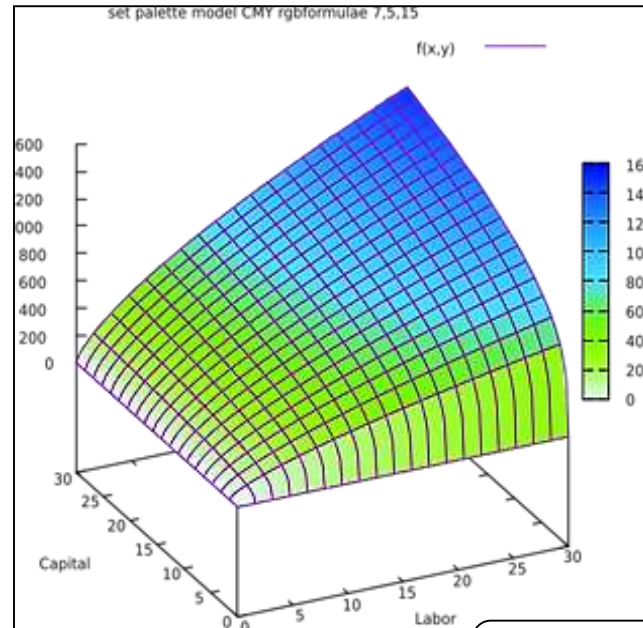
The Shapiro-Wilk W-statistic for the Skew model (0.98251) is much closer to 1, indicating **residuals are closer to a normal distribution**. Classical model residuals deviate strongly from normality (W = 0.77591).

The Cobb-Douglas Plot

Using the Coefficients as elasticity in form



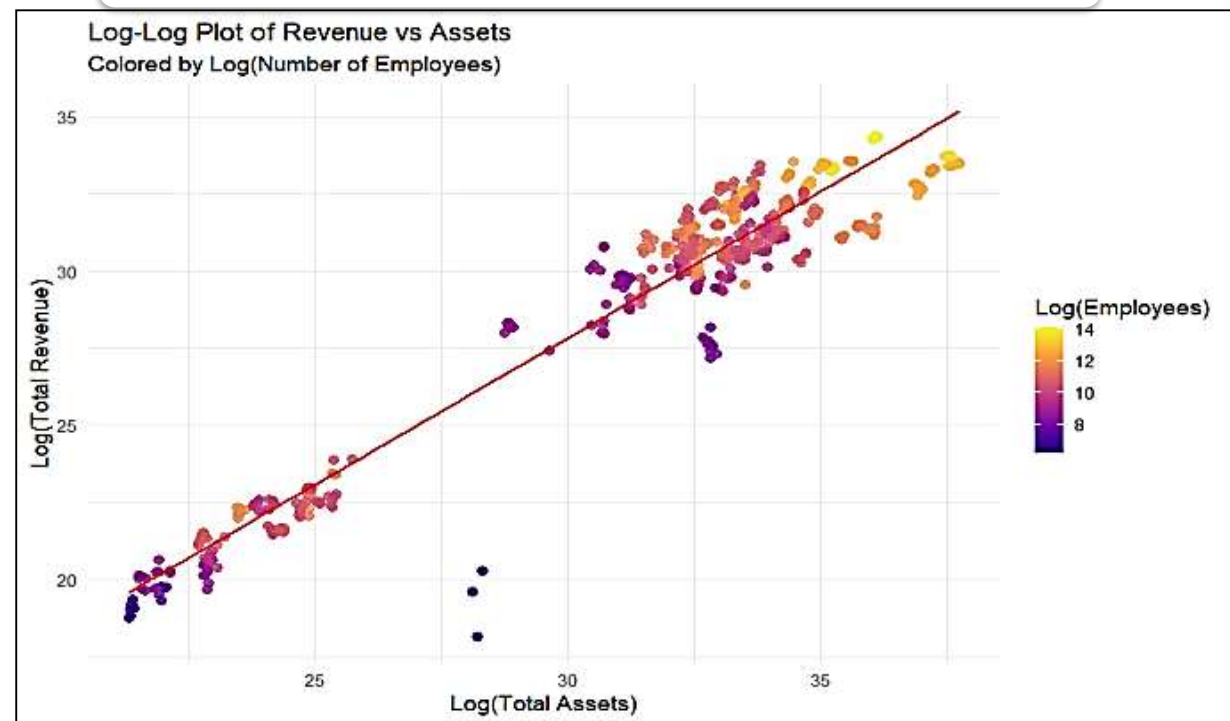
Typical Cobb Douglas



Source :
https://www.economi.ki.com/_media/en/cobb-douglas-1.svg.png

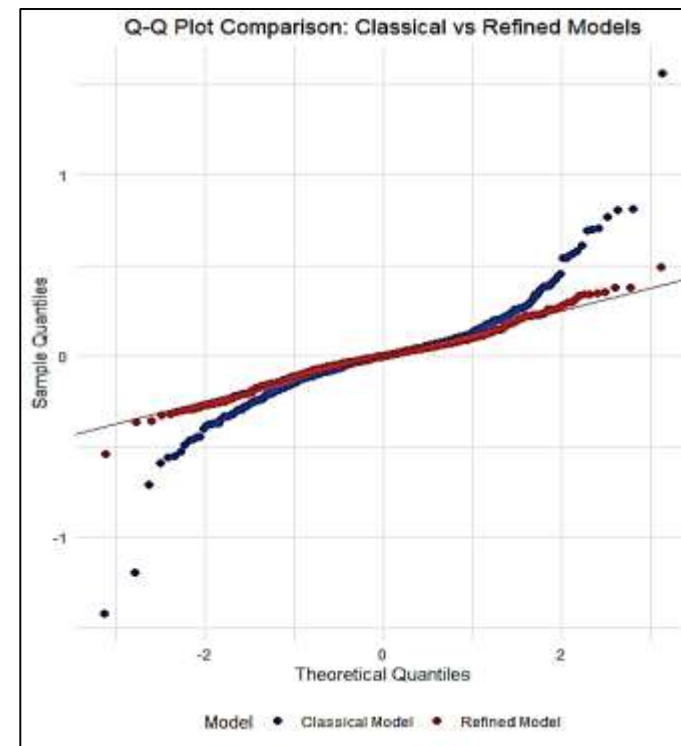
The empirical plot follows the Cobb-Douglas Production Function Log-Linear and Logarithmic exponential form

Linear Plot



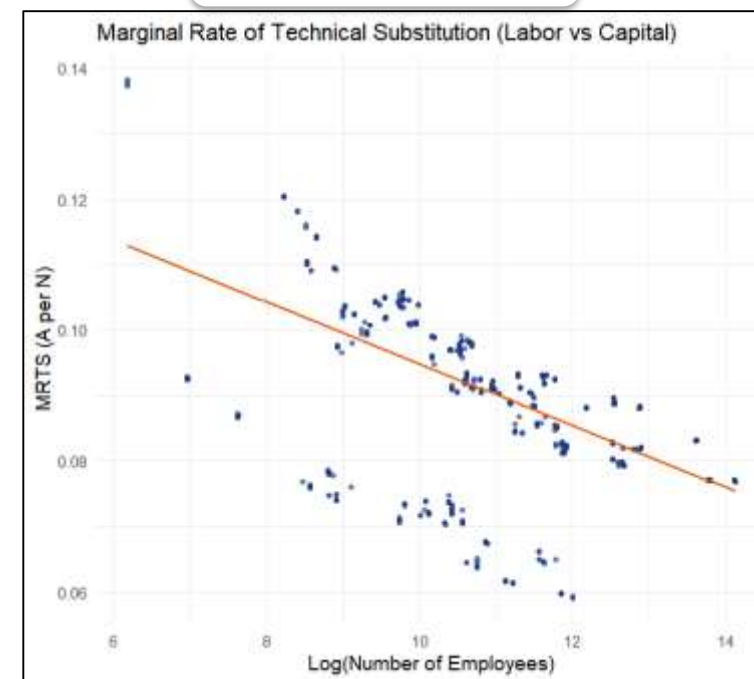
The plot shows a very linear good to fit alignment of Total Revenue with Total Assets whilst assuming the Number of Employees is Constant

QQ-Plot



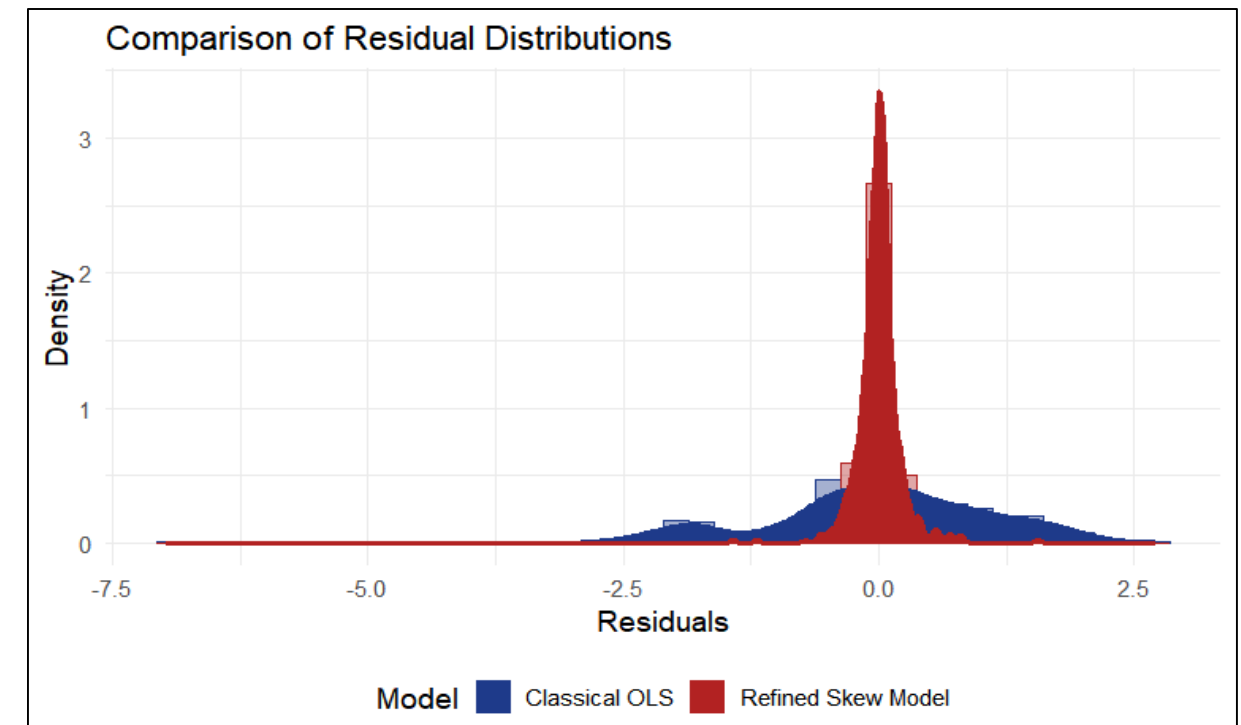
The model when plotted in R's Q-Q plot shows that the theoretical quantiles are almost linear for the Skew-Symmetric

MRTS



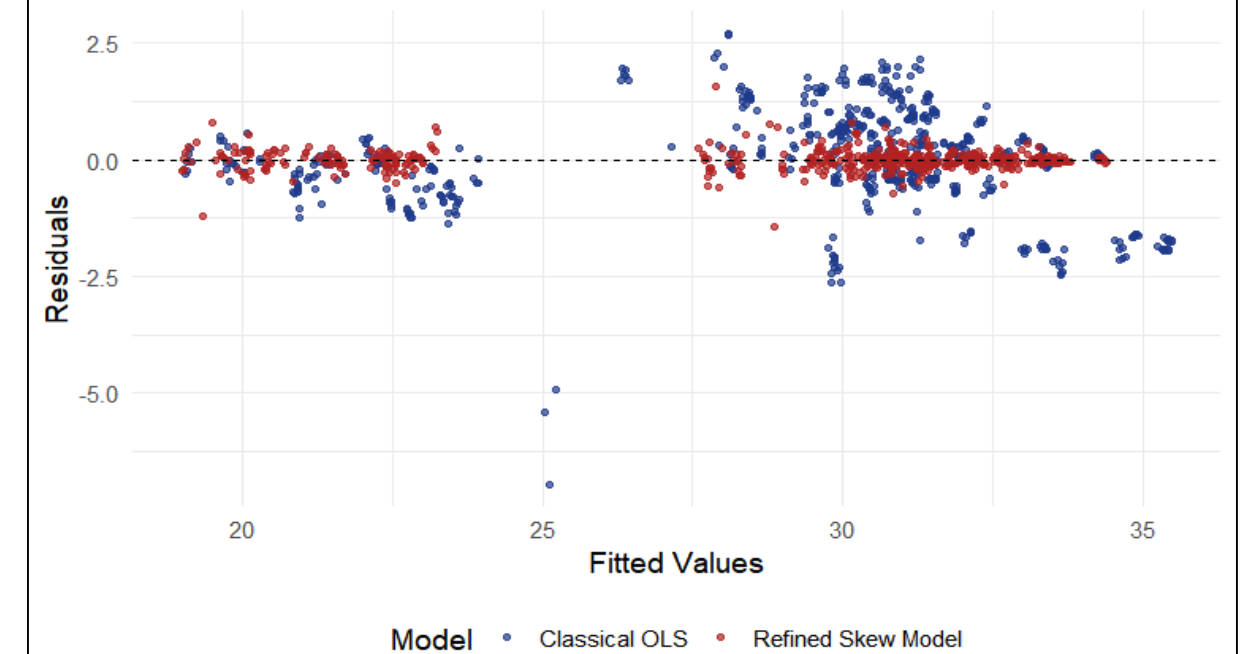
The MRTS scatter plot confirms that as firms become larger in terms of labor, each additional worker substitutes less capital

Residuals comparison



Under a skew-normal likelihood, the refined model centers the residuals and reduces dispersion, in which confirming that the skew-symmetric assumption better represents the underlying data-generating process

Residuals vs Fitted: Classical vs Refined Model



The skew-normal specification produces centered and homoscedastic residuals, indicating improved distributional symmetry and model fit relative to the OLS benchmark

4. Discussion and Limitations

Discussion and Interpretation

Prior Research Alignment

1.

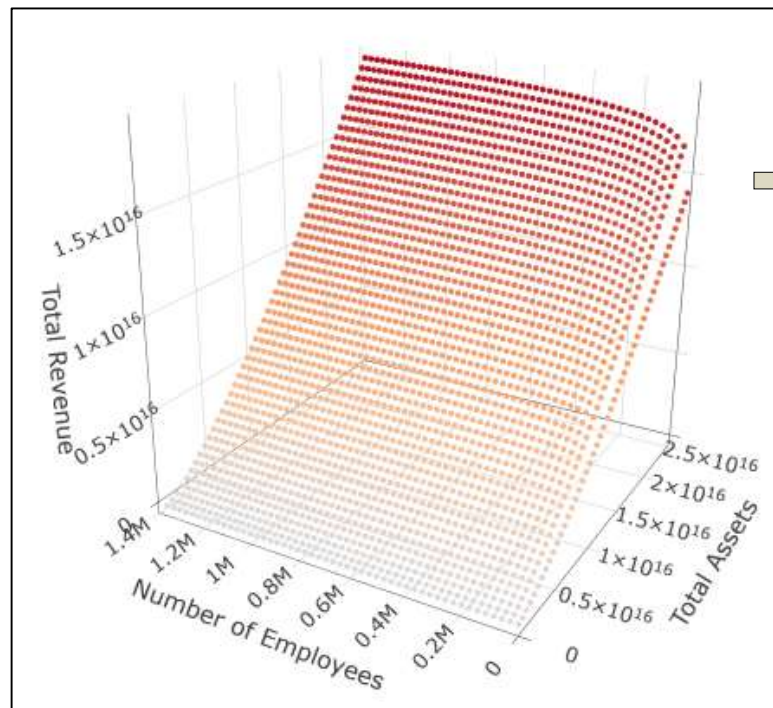
- Findings align with Jimichi et al.'s application of the Cobb–Douglas and skew-symmetric model in the Japanese market. [1]
- Confirms the model's reliability in capturing elasticity and asymmetry in firm-level financial data.

Structural Market Differences

2.

- Japan:** high capital intensity, aging demographics, and strong capital-labor substitution. [2]
- Indonesia:** shift from manufacturing to resource-based sectors increases aggregate capital intensity and heterogeneity. [3]

Referencing the previous graph...



- Indonesia's revenue surface driven by **asset growth**.
- Suggests a capital-intensive economy, where mining and resource firms dominate revenue generation patterns.

Comparison with our Method

3.

- Alternative econometric methods include:
 - SFA (Stochastic Frontier Analysis) for productivity/efficiency.
 - GMM & Panel Cointegration Tests (PCT) for macro-financial linkages.
- Present study's Cobb–Douglas + Skew-Symmetric model preserves interpretability and captures fat-tailed residuals typical in emerging markets.

- Jimichi, M., Kawasaki, Y., Miyamoto, D., Saka, C., & Nagata, S. (2023). Statistical Modelling of Financial Data with Skew-Symmetric Error Distributions. *Symmetry*, 15(9), 1772.
- International Monetary Fund. (2010). Japan's quest for growth: Exploring the role of capital and innovation (Working Paper No. 10/294). IMF.
- Rudolf, J. (2025, August 21). *Why a commodities boom is not lifting Indonesia's economy*. Financial Times.

Limitations

1.

Limited Time Span

- Dataset sourced from Yahoo Finance for simplicity.
- Limited time span (1–2 years) of quarterly data reduces robustness compared to multi-year studies.

2.

Conservative Modelling

- Intentional simplicity: regression with few core variables for clear log-likelihood optimization.
- Ensures interpretability but trades off robustness of richer variable inclusion.

3.

Comparison with Prior studies

- Jimichi's model included industry dummy variables and multiple explanatory factors.
- Our conservative model omits variables like profit ratios and sectoral adjustments, reducing generalization power.

Further Suggestions

1.

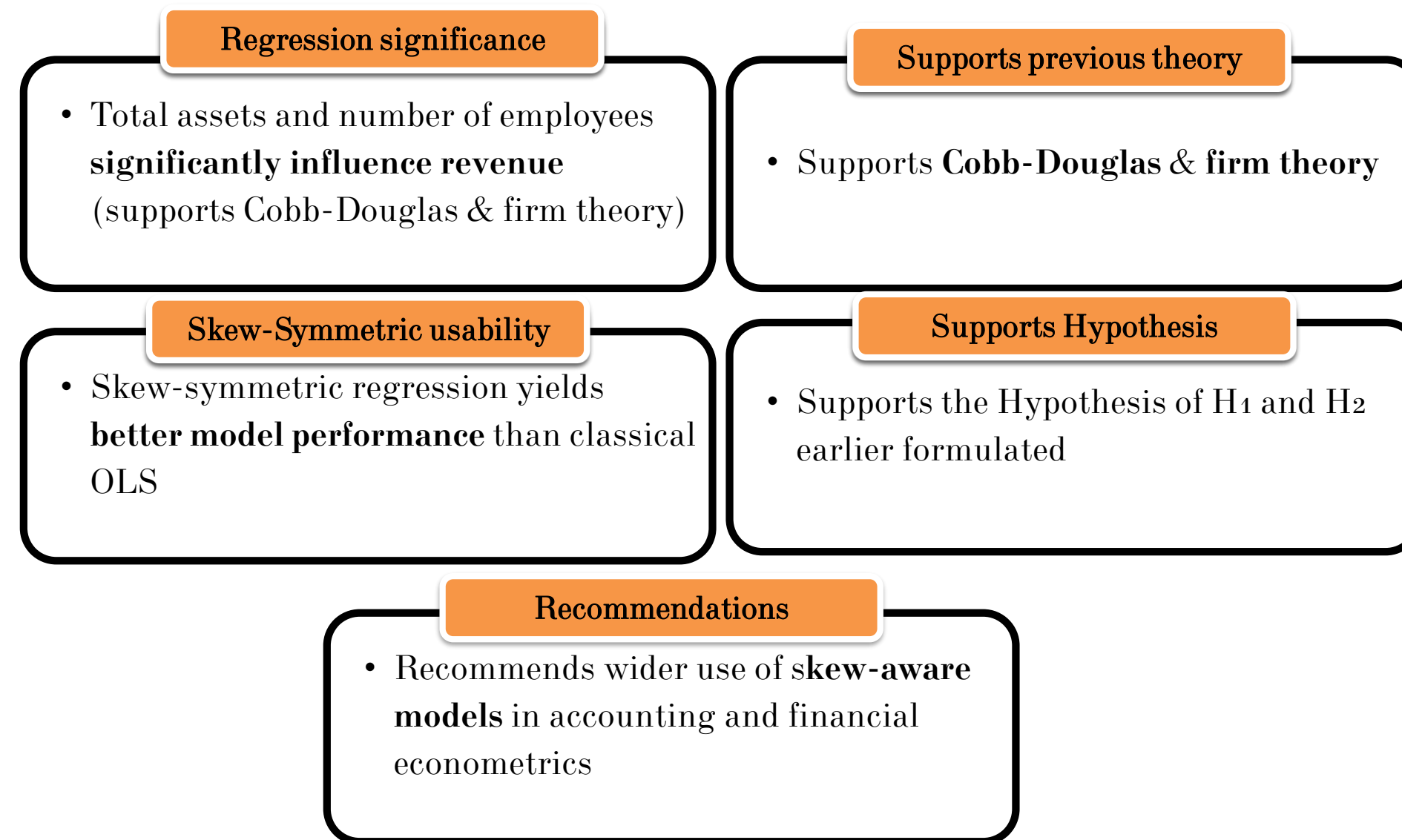
Extend dataset across more years and firms

2.

Incorporate comprehensive financial statement variables

3.

Explore non-conservative formulations



Thank you